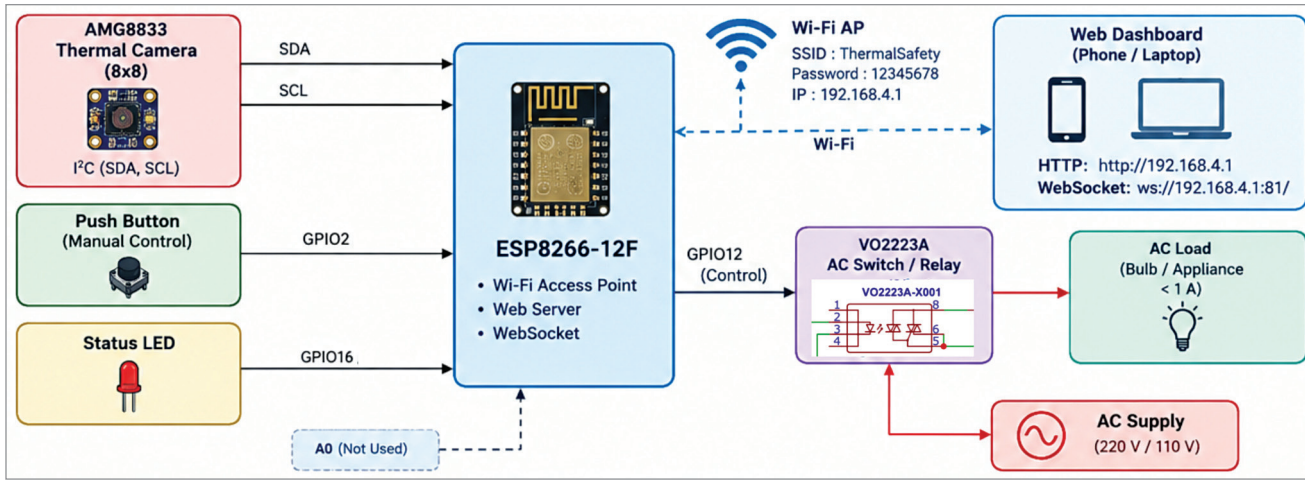


Fig. 2: Block diagram of the system

Properly calibrated, the switch makes overheating detection more reliable. Fig. 1 shows the author's prototype, while Fig. 2 depicts the block diagram of the system.

What will you learn?

By building this project, readers will learn how to capture thermal images using an I²C-based Grid-EYE sensor, implement digital input/output control for load switching, establish Wi-Fi connectivity, and develop a real-time web application using WebSockets. The project also demonstrates power regulation, opto-isolated AC load control, and the integration of thermal sensing, processing, switching, and remote monitoring into a compact system.

Parts list and cost

The design is built around the ESP-12F Wi-Fi module and the VO2223A-X001 opto-TRIAC for isolated AC load control. The components, their designators, and estimated prices are listed in the Bill of Materials table.

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